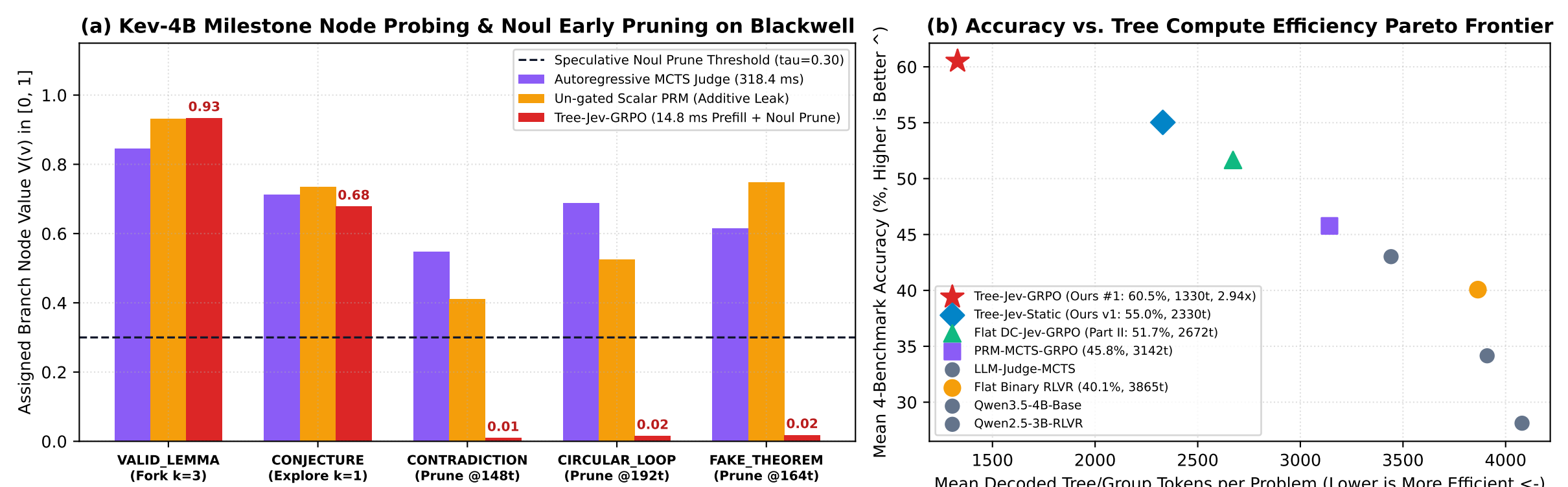
(a) Kev-4B Milestone Node Probing & Noul Early Pruning on Blackwell
Assigned Branch Node Value V(v) in [0, 1]
Speculative Noul Prune Threshold (tau=0.30)
Autoregressive MCTS Judge (318.4 ms)
Un-gated Scalar PRM (Additive Leak)
Tree-Jev-GRPO (14.8 ms Prefill + Noul Prune)
0.93
0.68
0.01
0.02
0.02
VALID_LEMMA (Fork k=3)
CONJECTURE (Explore k=1)
CONTRADICTION (Prune @148t)
CIRCULAR_LOOP (Prune @192t)
FAKE_THEOREM (Prune @164t)
(b) Accuracy vs. Tree Compute Efficiency Pareto Frontier
Mean 4-Benchmark Accuracy (%, Higher is Better ^)
Mean Decoded Tree/Group Tokens per Problem (Lower is More Efficient <-)
Tree-Jev-GRPO (Ours #1: 60.5%, 1330t, 2.94x)
Tree-Jev-Static (Ours v1: 55.0%, 2330t)
Flat DC-Jev-GRPO (Part II: 51.7%, 2672t)
PRM-MCTS-GRPO (45.8%, 3142t)
LLM-Judge-MCTS
Flat Binary RLVR (40.1%, 3865t)
Qwen3.5-4B-Base
Qwen2.5-3B-RLVR